

CS 173 — Diagrams & Math Reference Guide

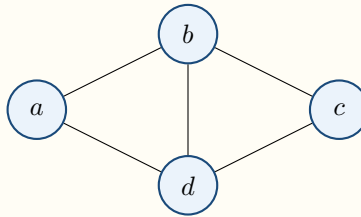
TikZ, proof typesetting, and CS173 notation

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Contents

Graph Theory Diagrams

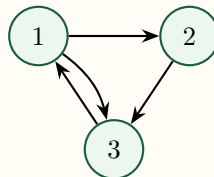
Simple undirected graph



```
\begin{tikzpicture}[
  vertex/.style={circle, draw=defblue, fill=boxbluebg, thick,
    minimum size=22pt, font=\small\bfseries},
]
\node[vertex] (a) at (0,0) {$a$};
\node[vertex] (b) at (2,1) {$b$};
\node[vertex] (c) at (4,0) {$c$};
\node[vertex] (d) at (2,-1) {$d$};

\draw (a) -- (b) -- (c) -- (d) -- (a);
\draw (b) -- (d);
\end{tikzpicture}
```

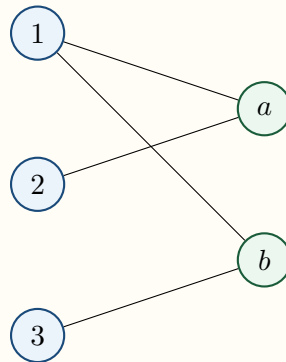
Directed graph (digraph)



```
\begin{tikzpicture}[
  vertex/.style={circle, draw=thmgreen, fill=boxgreenbg, thick,
    minimum size=22pt},
  ->, >=Stealth, thick,
]
\node[vertex] (1) at (0,0) {$1$};
\node[vertex] (2) at (2,0) {$2$};
\node[vertex] (3) at (1,-1.5) {$3$};

\draw (1) -- (2);
\draw (2) -- (3);
\draw (3) -- (1);
\draw (1) edge[bend left=20] (3); % curved arrow
\end{tikzpicture}
```

Bipartite graph



```
% Two node styles for the two parts
\node[L] (l1) at (0, 2) {$1$}; \node[L] (l2) at (0, 0) {$2$};
\node[R] (ra) at (3, 1) {$a$}; \node[R] (rb) at (3,-1) {$b$};
\draw (l1)--(ra); \draw (l1)--(rb); \draw (l2)--(ra);
```

Edge labels



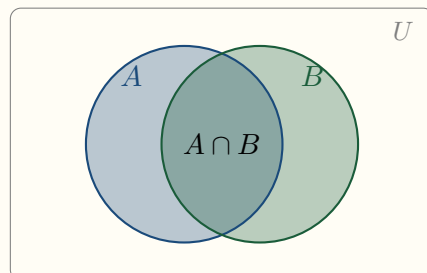
```
\draw (u) -- node[above, font=\small]{weight 5$} (v);
```

Self-loops

```
\draw (v) edge[loop above] node{} (v);
```

Set Diagrams (Venn Diagrams)

Two-set Venn diagram



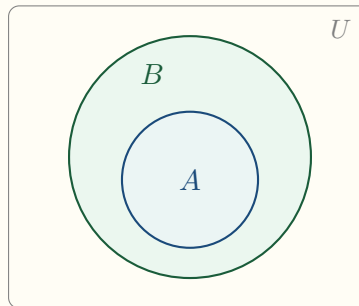
```
\def\radius{1.3} \def\overlap{1.0}
\begin{scope}[fill opacity=0.3]
  \fill[defblue] (-\overlap/2, 0) circle (\radius);
  \fill[thmgreen] (\overlap/2, 0) circle (\radius);
\end{scope}
```

```

\end{scope}
\draw[defblue, thick] (-\overlap/2, 0) circle (\radius);
\draw[thmgreen, thick] ( \overlap/2, 0) circle (\radius);
\node at (-\overlap/2 - 0.7, 0.9) {\color{defblue}$A$};
\node at ( \overlap/2 + 0.7, 0.9) {\color{thmgreen}$B$};

```

Subset diagram ($A \subseteq B$)



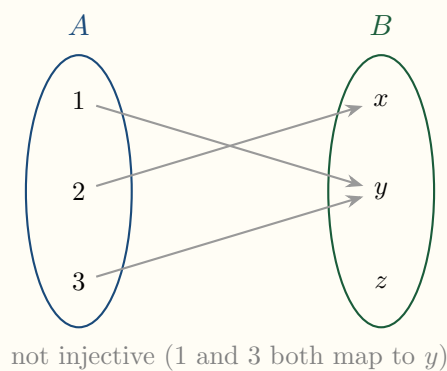
```

% Outer circle = B, inner circle = A
\draw[thmgreen, thick, fill=boxgreenbg] (0,0) circle (1.6);
\draw[defblue, thick, fill=boxbluebg, fill opacity=0.5] (0,-0.3) circle (0.9);
\node at (0,-0.3) {\color{defblue}$A$};
\node at (-0.5,1.1) {\color{thmgreen}$B$};

```

Function Diagrams

Arrow diagram (mapping between finite sets)



```

% Draw domain ellipse
\draw[defblue, thick] (0,0) ellipse (0.7 and 1.8);
\node (a1) at (0, 1.2) {$1$}; \node (a2) at (0,0) {$2$};

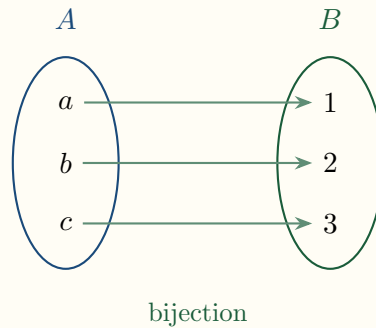
% Draw codomain ellipse
\draw[thmgreen, thick] (4,0) ellipse (0.7 and 1.8);
\node (b1) at (4, 1.2) {$x$}; \node (b2) at (4,0) {$y$};

% Arrows

```

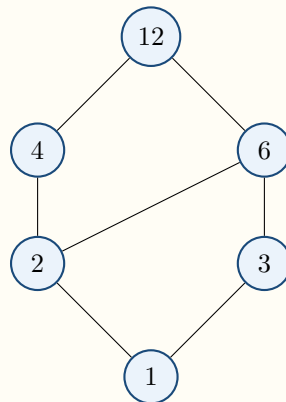
```
\draw[->, >=Stealth, gray!80] (a1) -- (b2);
\draw[->, >=Stealth, gray!80] (a2) -- (b1);
```

Bijection diagram



Hasse Diagrams (Partial Orders)

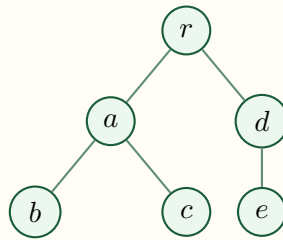
Hasse diagrams omit arrows (direction is up) and transitive edges.



```
% Divisibility poset on {1,2,3,4,6,12} only cover relations drawn
\node[node] (1) at (0,0) {$1$};
\node[node] (2) at (-1.5,1.5) {$2$}; \node[node] (3) at (1.5,1.5) {$3$};
\node[node] (4) at (-1.5,3) {$4$}; \node[node] (6) at (1.5,3) {$6$};
\node[node] (12) at (0,4.5) {$12$};
\draw (1)--(2); \draw (1)--(3); \draw (2)--(4); \draw (2)--(6);
\draw (3)--(6); \draw (4)--(12); \draw (6)--(12);
```

Trees (Induction / Graph Theory)

Binary tree



```
\usepackage{tikz} % child/sibling syntax is built in

\begin{tikzpicture}[level distance=1.2cm, sibling distance=2.0cm]
  \node[n] {$r$}
    child { node[n] {$a$}
      child { node[n] {$b$} }
      child { node[n] {$c$} }
    }
    child { node[n] {$d$}
      child { node[n] {$e$} }
    };
\end{tikzpicture}
```

Labelling leaf nodes differently

```
% Define two node styles
\tikzstyle{inner} = [circle, draw=thmgreen, fill=boxgreenbg, thick, minimum size=18pt]
\tikzstyle{leaf} = [circle, draw=warnAmber, fill=boxamberbg, thick, minimum size=18pt]

\node[inner] {$r$}
  child { node[inner] {$v$}
    child { node[leaf] {$\ell_1$} }
    child { node[leaf] {$\ell_2$} }
  };
```

Induction Proof Structure

Use the description environment for the standard three-part layout.

```
\begin{proof}[Proof by (strong) induction on $n$]
  \begin{description}[leftmargin=2cm, labelwidth=2cm]

    \item[Base case.] ($n = 0$) ...

    \item[IH.] Assume $P(k)$ holds for all $k < n$.

    \item[IS.] We show $P(n)$ ...

  \end{description}
\end{proof}
```

Rendered output:

Proof by strong induction on n . **Base case.** ($n = 0$) $P(0)$ holds because ...

IH. Assume $P(k)$ holds for all $0 \leq k < n$.

IS. We show $P(n)$...

□

Aligned Proof Steps with Justifications

Use `align` or `alignat` to line up equals signs and add reasons.

```
\begin{align*}
|A \cup B| &= |A| + |B| - |A \cap B| && \text{(inclusion-exclusion)} \\
&= 5 + 7 - 3 && \text{(given)} \\
&= 9.
\end{align*}
```

$$\begin{aligned} |A \cup B| &= |A| + |B| - |A \cap B| && \text{(inclusion-exclusion)} \\ &= 5 + 7 - 3 && \text{(given)} \\ &= 9. \end{aligned}$$

Cases

```
\begin{proof}
We consider two cases.
\begin{description}
\item[Case 1:  $n$  even.] Write  $n = 2k$ . Then \ldots
\item[Case 2:  $n$  odd.] Write  $n = 2k+1$ . Then \ldots
\end{description}
\end{proof}
```

Truth Tables

P	Q	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$
T	T	T	T	T
T	F	F	T	F
F	T	F	T	T
F	F	F	F	T

```
\begin{tabular}{c c | c | c | c}
$P$ & $Q$ & $P \land Q$ & $P \lor Q$ & $P \Rightarrow Q$ \\
\hline
T & T & T & T & T \\
T & F & F & T & F \\
F & T & F & T & T \\
F & F & F & F & T
\end{tabular}
```

Counting / Combinatorics

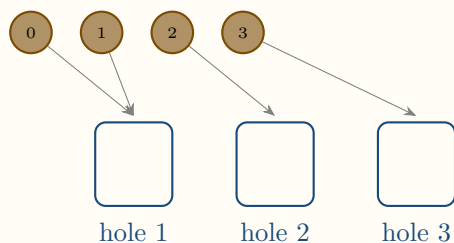
Binomial coefficients

```
\binom{n}{k} % C(n,k)
```

Permutations / products

```
\prod_{i=1}^n a_i  
\sum_{k=0}^n \binom{n}{k}
```

Pigeonhole illustration



Cardinality & Cantor Diagrams

Diagonal argument schematic

$$f(1) = \boxed{0}, 1, 0, \dots$$

$$f(2) = 1, 0, \boxed{1}, \dots$$

$$f(3) = 0, 0, 1, 1, \boxed{0}$$

$\vdots$

flip diagonal $\Rightarrow$ new sequence not in list

CS 173 Math Symbol Quick Reference

Meaning	Command	Output
Natural numbers	<code>\mathbb{N}</code>	$\mathbb{N}$
Integers	<code>\mathbb{Z}</code>	$\mathbb{Z}$
Reals	<code>\mathbb{R}</code>	$\mathbb{R}$
Power set	<code>\mathcal{P}(A)</code>	$\mathcal{P}(A)$
Set-builder	<code>\{x \mid P(x)\}</code>	$\{x \mid P(x)\}$
Cardinality	<code>\lvert A \rvert</code>	$ A $
Divides	<code>a \mid b</code>	$a \mid b$
Does not divide	<code>a \nmid b</code>	$a \nmid b$
Modulo	<code>a \equiv b \pmod{n}</code>	$a \equiv b \pmod{n}$
For all	<code>\forall</code>	$\forall$
There exists	<code>\exists</code>	$\exists$
Implies	<code>\Rightarrow</code>	$\Rightarrow$
Iff	<code>\iff</code>	$\iff$
Negation	<code>\neg</code>	$\neg$
Conjunction	<code>\land</code>	$\wedge$
Disjunction	<code>\lor</code>	$\vee$
Subset	<code>\subseteq</code>	$\subseteq$
Proper subset	<code>\subsetneq</code>	$\subsetneq$
Intersection	<code>\cap</code>	$\cap$
Union	<code>\cup</code>	$\cup$
Set difference	<code>A \setminus B</code>	$A \setminus B$
Empty set	<code>\emptyset</code>	$\emptyset$
Contradiction	<code>\Rightarrow \Leftarrow</code>	$\Rightarrow \Leftarrow$
Floor	<code>\lfloor x \rfloor</code>	$\lfloor x \rfloor$
Ceiling	<code>\lceil x \rceil</code>	$\lceil x \rceil$
Composition	<code>f \circ g</code>	$f \circ g$

Useful TikZ Tips

Relative node placement — use positioning library:

```
\usetikzlibrary{positioning}
\node (a) {$a$};
\node[right=1.5cm of a] (b) {$b$}; % 1.5cm to the right of a
\node[below=of b] (c) {$c$}; % default gap below b
```

Foreach loops — draw grids, regular polygons, etc.:

```
\foreach \i in {0,60,...,300} {
  \node[circle, fill=defblue!30] at (\i:1.5) {};
}
```

Curly brace annotation — label a span of nodes:

```
\usetikzlibrary{decorations.pathreplacing}
\draw[decorate, decoration={brace, amplitude=6pt}]
  (0,0) -- (3,0) node[midway, below=8pt]{$n$ elements};
```

Wrapping TikZ in a figure:

```
\begin{figure}[h]
  \centering
  \begin{tikzpicture} ... \end{tikzpicture}
  \caption{A graph on 4 vertices.}
  \label{fig:mygraph}
\end{figure}
% Reference with \ref{fig:mygraph}
```